

Terraform script to deploy an EC2 instance with CloudWatch monitoring enabled

Pre Requisite : Terraform and AWS CLI

```
.....

provider "aws" {
  region = "us-east-2"
}

# Generate a new SSH key pair
resource "tls_private_key" "ec2_key" {
  algorithm = "RSA"
  rsa_bits  = 4096
}

resource "aws_key_pair" "ec2_key" {
  key_name   = "ec2-key"
  public_key = tls_private_key.ec2_key.public_key_openssh
}

resource "local_file" "private_key" {
  content      = tls_private_key.ec2_key.private_key_pem
  filename     = "ec2-key.pem"
  file_permission = "0400"
}

# Security Group: allow SSH and HTTP
resource "aws_security_group" "ec2_sg" {
  name        = "ec2-sg"
  description = "Allow SSH and HTTP"

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
```

```
}
```

```
# IAM Role for CloudWatch Agent
```

```
resource "aws_iam_role" "cw_agent_role" {  
  name = "cloudwatch-agent-role"
```

```
  assume_role_policy = jsonencode({  
    Version = "2012-10-17",  
    Statement = [{  
      Effect = "Allow",  
      Principal = {  
        Service = "ec2.amazonaws.com"  
      },  
      Action = "sts:AssumeRole"  
    }]  
  })  
}
```

```
resource "aws_iam_role_policy_attachment" "cw_attach" {  
  role      = aws_iam_role.cw_agent_role.name  
  policy_arn = "arn:aws:iam::aws:policy/CloudWatchAgentServerPolicy"  
}
```

```
# Instance profile
```

```
resource "aws_iam_instance_profile" "cw_instance_profile" {  
  name = "cloudwatch-agent-profile"  
  role = aws_iam_role.cw_agent_role.name  
}
```

```
# EC2 Instance with CloudWatch Agent setup
```

```
resource "aws_instance" "example" {  
  ami              = "ami-0cb91c7de36eed2cb" # Ubuntu 22.04 LTS in us-east-2  
  instance_type    = "t2.micro"  
  key_name          = aws_key_pair.ec2_key.key_name  
  monitoring        = true  
  vpc_security_group_ids = [aws_security_group.ec2_sg.id]  
  iam_instance_profile = aws_iam_instance_profile.cw_instance_profile.name
```

```
  user_data = <<-EOF  
    #!/bin/bash  
    sudo apt update  
    sudo apt install -y awscli unzip
```

```
    # Download and install CloudWatch Agent  
    curl https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/amazon-  
cloudwatch-agent.deb -o amazon-cloudwatch-agent.deb  
    sudo dpkg -i -E ./amazon-cloudwatch-agent.deb
```

```
    # Create config directory  
    sudo mkdir -p /opt/aws/amazon-cloudwatch-agent/etc
```

```
    # Create CloudWatch config file  
    cat <<EOC > /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json  
    {  
      "agent": {  
        "metrics_collection_interval": 60,  
        "run_as_user": "root"
```

```

    },
    "metrics": {
      "append_dimensions": {
        "InstanceId": "${aws:InstanceId}"
      },
      "metrics_collected": {
        "cpu": {
          "measurement": [
            "cpu_usage_idle",
            "cpu_usage_iowait",
            "cpu_usage_user"
          ],
          "metrics_collection_interval": 60
        },
        "mem": {
          "measurement": [
            "mem_used_percent"
          ],
          "metrics_collection_interval": 60
        }
      }
    }
  }
}
EOC

```

```

# Start the CloudWatch Agent
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
  -a fetch-config -m ec2 \
  -c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -s
EOF

```

```

tags = {
  Name = "cloudwatch-monitored-instance"
}
}

```

```

terraform init
terraform plan
terraform validate
terraform apply -auto-approve

```

In the AWS Console, go to:

CloudWatch → Metrics → All Metrics → CWAgent

You'll see mem_used_percent, cpu_usage_user, etc.

